

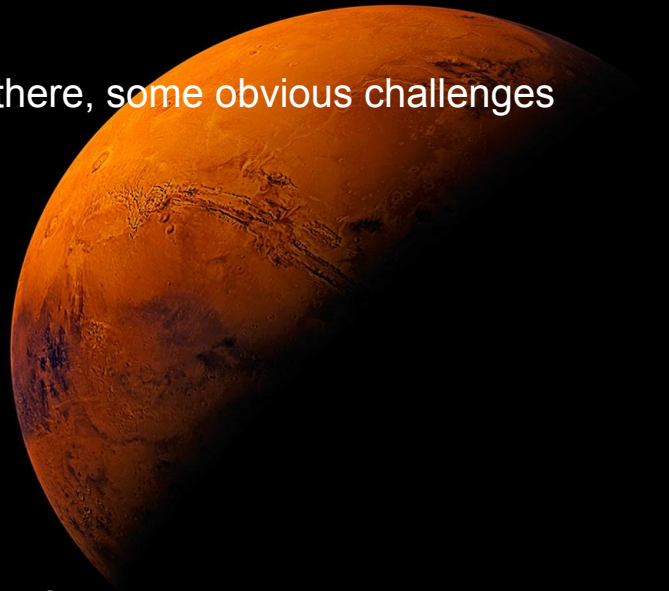
CL - 246
Final Presentation
G24

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Somewhere in the Future, humans will make it to Mars (thanks Elon)
Unfortunately this rogue planet is nothing like Earth

On the path to inhabit humans there, some obvious challenges arise-

- Extreme temperatures
- Radiation
- Different atmosphere
- Gravity
- Loneliness



For this project we looked at the first two pain points exclusively

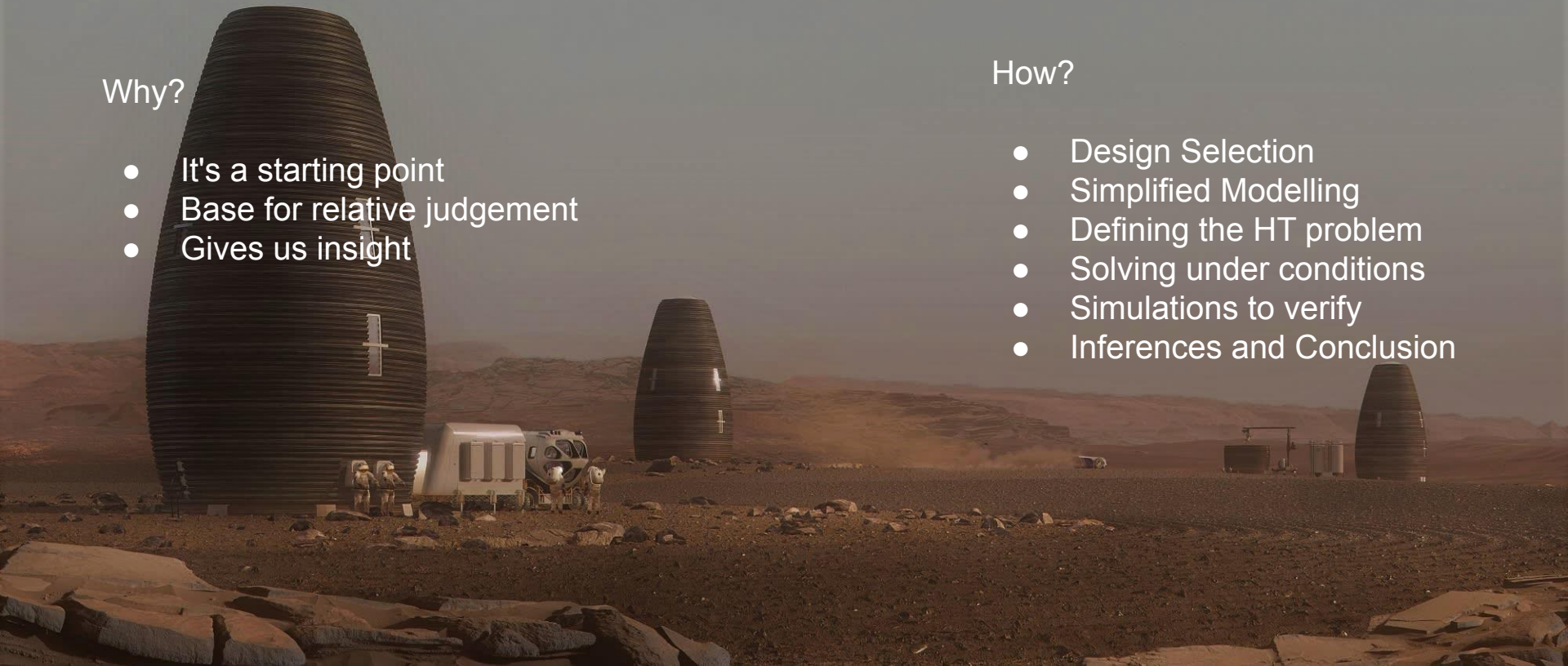
We took our existing design of houses and studied them under martian conditions

Why?

- It's a starting point
- Base for relative judgement
- Gives us insight

How?

- Design Selection
- Simplified Modelling
- Defining the HT problem
- Solving under conditions
- Simulations to verify
- Inferences and Conclusion



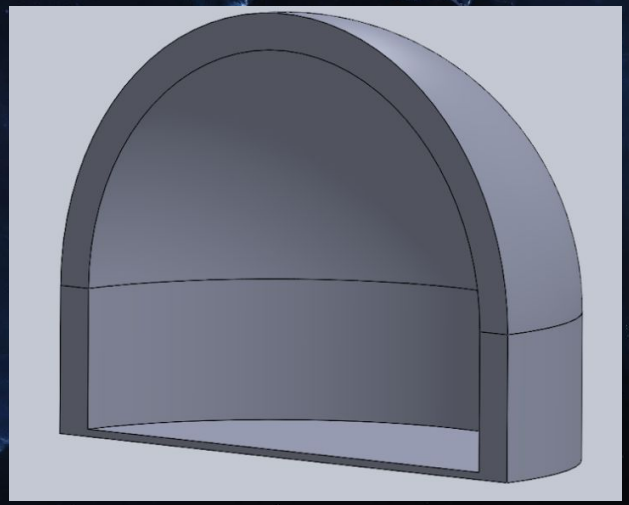
Design Selection & Modelling

Looked at colder regions of Earth, came across DOME HOMES

Why?

- Energy efficient
- Lower cost of building
- Disaster Prone





Solidworks

Heat Transfer Problem

Consider the dome shaped (hemisphere above cylinder) house situated on Mars with certain size and a constant heat source inside maintaining the temperature. With suitable assumptions and under certain conditions calculate the following :-

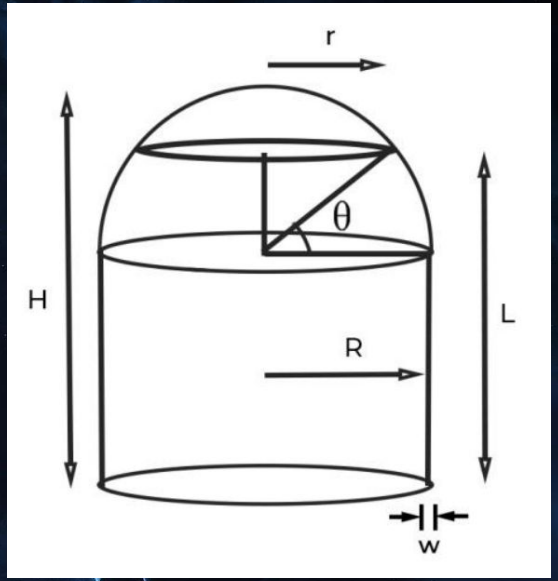
- Convective heat transfer coefficient between outer walls of the structure and atmosphere
- Temperature of outer wall of structure
- Total heat loss from the structure
- Average cost of heating per day

Conditions

- Atmospheric Temperature: -60°C
- Average wind velocity : 5 ms^{-1}

Assumptions

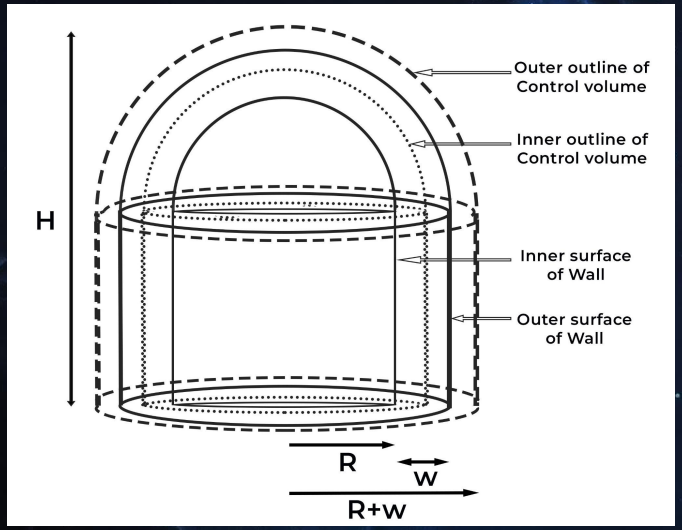
- Fixed Inner Wall Temperature: 22°C
- Total Heat Lost = Total Heat Generated
- Atmosphere consists of only CO_2 (extrapolated)

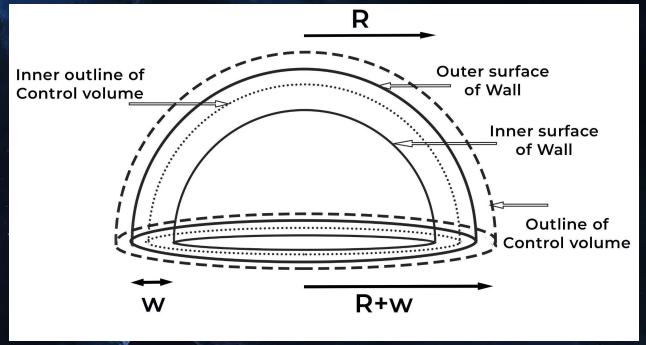


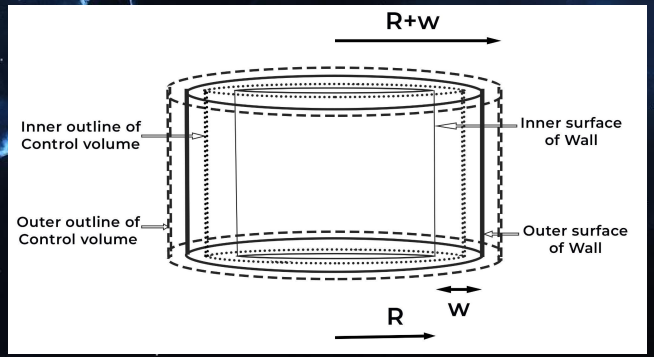
Structure Specifications

- R = 2 m
- w = 0.5 m
- H - R = 3m
- Glass

Control Volume

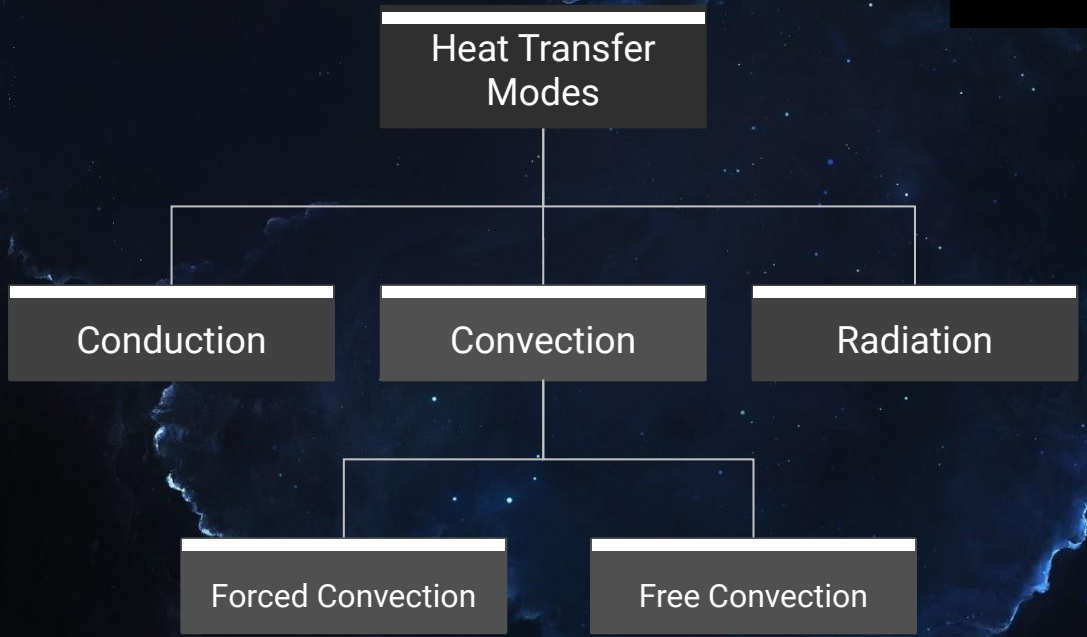


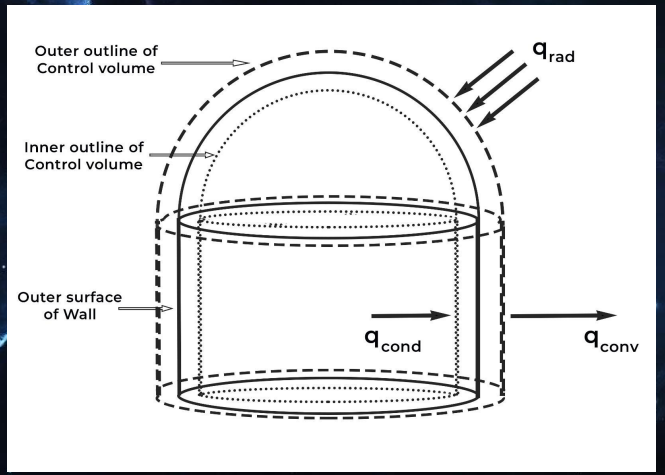




Solution Overview

$$Q_{conduction} + Q_{solar-radiation} = Q_{convection}$$





Terminologies for Convection

Reynolds Number

Ratio of inertial forces to viscous forces within a fluid

Rayleigh Number

Dimensionless **number** associated with buoyancy-driven flow which characterises fluid flow regime

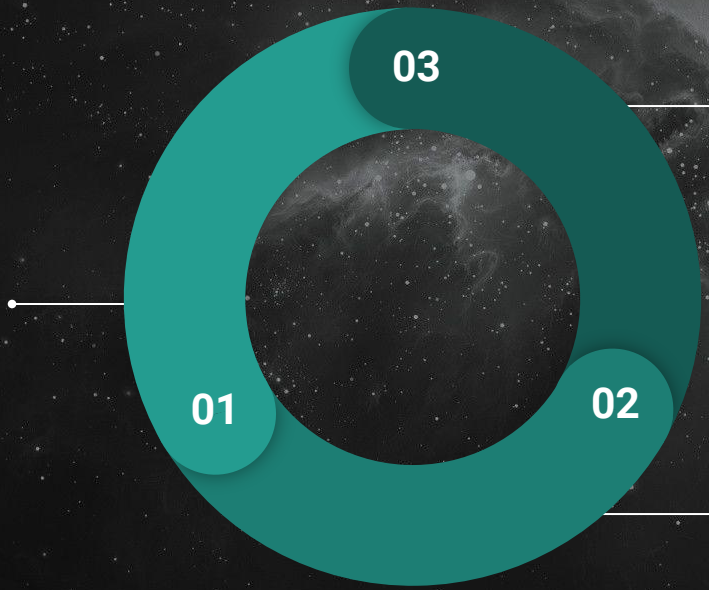
Nusselt Number

Ratio of convective to conductive heat transfer across a boundary.

Convection Process

Calculating Reynolds/Rayleigh Number

Factors used to decide appropriate Nusselt Number Correlation



Calculating Convective HT Coefficient

$$h = \frac{Nu * k}{L}$$

Calculating Nusselt Number

Plugging values in the particular correlation

Convection (Cylinder)

Free Convection

$$Ra_L = \frac{g\beta L^3 (T_{s,cylinder} - T_\infty)}{\nu\alpha}$$

$$\overline{Nu}_{CFree} = \left\{ 0.825 + \frac{0.387 Ra_L^{\frac{1}{6}}}{\left[1 + \left(\frac{0.492}{Pr}\right)^{\frac{9}{16}}\right]^{\frac{8}{27}}} \right\}^2$$

$$h_{CFree} = \frac{Nu_{CFree} k}{L}$$

h_CFree = 4.0466 W / m² K

Forced Convection

$$Re_{CFor} = \frac{VD}{\nu}$$

$$\overline{Nu}_{CFor} = 0.3 + \frac{0.62\,\mathrm{Re}_D^{1/2}\,\mathrm{Pr}^{1/3}}{\left[1 + (0.4/Pr)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{Re_D}{282,000}\right)^{5/8}\right]^{4/5}$$

$$h_{CFor} = \frac{Nu_{CFor} k}{L}$$

h_Cfor = 12.9843 W / m² K

Convection (Dome)

Free Convection

$$Ra_{DFree} = \frac{g\beta\ D^3\left(T_{s,dome} - T_\infty\right)}{\nu\alpha}$$

$$\overline{Nu}_{DFree} = 2 + \frac{0.589 Ra_D^{1/4}}{\left[1 + (0.469/\Pr)^{9/16}\right]^{4/9}}$$

$$h_{DFree} = \frac{Nu_{DFree} k}{L}$$

h_DFree = 1.8722 W / m² K

Forced Convection

$$Re_{DFor} = \frac{VD}{\nu}$$

$$\overline{Nu}_{DFor} = 2 + \left(0.4 Re_D^{1/2} + 0.06 Re_D^{2/3}\right) \Pr^{0.4} \left(\frac{\mu}{\mu_s}\right)^{1/4}$$

$$h_{DFor} = \frac{Nu_{DFor} \cdot k}{D}$$

h_DFor = 6.1843 W / m² K

Convection Results

The net heat transfer coefficient for dome and cylinder are:-

- h_C = 13.1227 W / m² K
- h_ D = 6.2396 W / m² K

$$h = (h_{Free}^3 + h_{For}^3)^{\frac{1}{3}}$$

Convection Equation

$$Q_{conv} = h * (\text{Surface Area}) * \Delta T$$

Solar Radiation

Variation in Solar radiation is governed by:

1. The Mars-Sun distance
2. Solar zenith angle
3. Opacity of the Martian atmosphere

The direct beam irradiance on Mars surface (Gb) is calculated using Beer's law

$$G_b = G_{ob} exp[\tau m(z)]$$

where τ is optical depth,
where m(z) is the air mass determined by the zenith angle z

$$m(z) = \frac{1}{cosz}$$

Solar Radiation

- Solar irradiance(G□) on surface is sum of direct(G_{bh}) and diffuse irradiance(G_{dh}) on Mars horizontal surface

$$G_h = G_{ob} cosz \frac{f(z,\tau)}{0.9}$$

$$G_{bh} = G_{ob} cosz exp(\frac{-\tau}{cosz})$$

Hourly radiation is given by:

$$I_{bh} = \frac{12^*}{\pi} G_{ob} \int_{\omega_1}^{\omega_2} (\sin\phi \sin\delta + \cos\phi \cos\delta \cos\omega) \exp[-\tau/(\sin\phi \sin\delta + \cos\phi \cos\delta \cos\omega)] d\omega$$

ϕ latitude

δ declination angle

ω hour angle measured from the true noon westward

It comes out to be 160 W/m^2

Conduction

- 1-D Steady State Conduction within walls
- Flow - From inner wall to outer wall
- Thermal Conductivity - SiO_2

$$Q_{cond} = k * w * \Delta T$$

Final Equation & Results

- Unknown is T_s
- Solved iteratively
- Convergence Criteria: $\Delta T_8 < 0.01$

$$Q_{conduction} + Q_{solar-radiation} = Q_{convection}$$

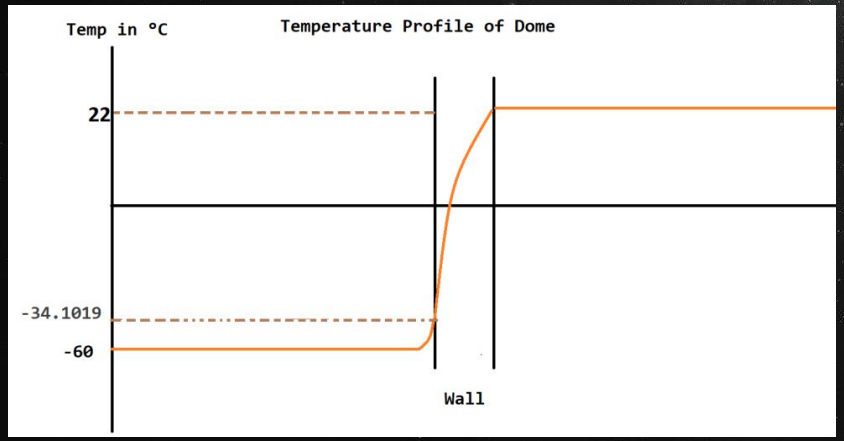
$$kw(T_i - T_s) + 160A = hA(T_s - T_a)$$

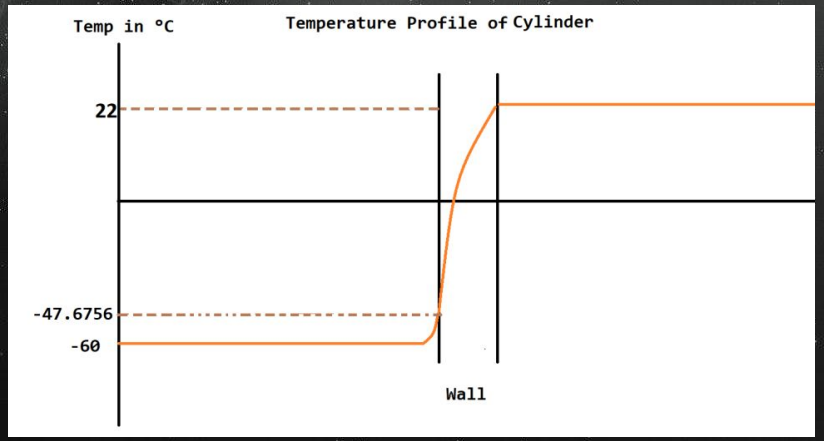
$$T_s = \frac{kwT_i + hAT_a + 160A}{hA + kw}$$

- T_8 (Cyc) = −47.739°C
- T_8 (Dome) = −34.219°C
- Q_loss (Cyc) = 160.888 W / m²
- Q_loss (Dome) = 160.859 W / m²

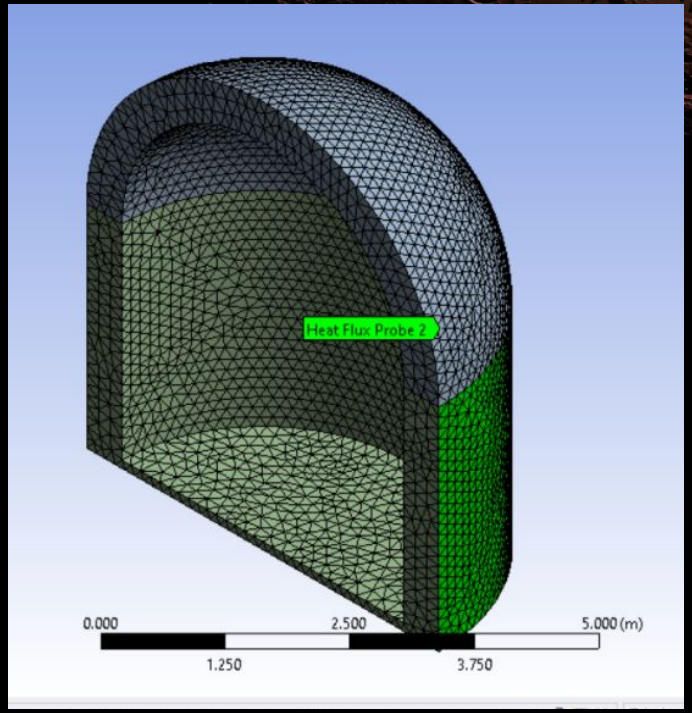
Temperature Profile

- ΔT (Dome) > ΔT (Cyc)
- Non-linear due to Radiation





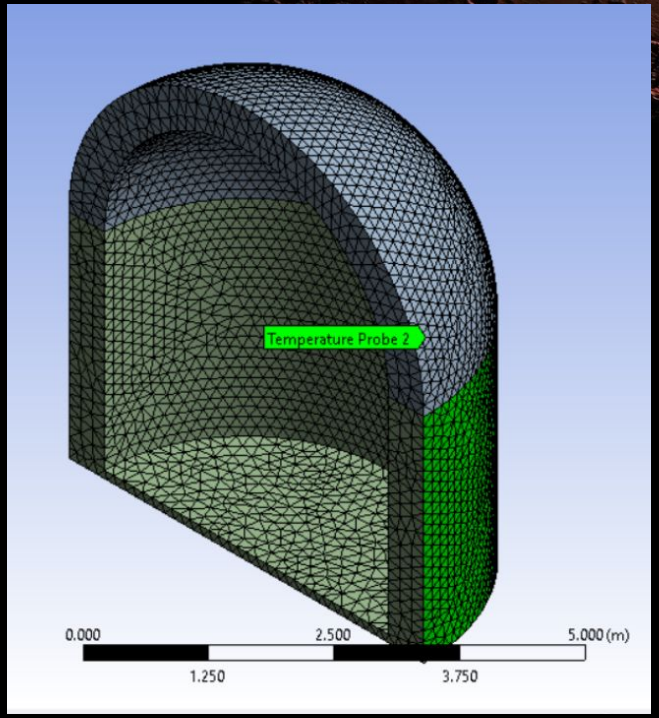
Simulations Results (Cyc1)



Details of "Heat Flux Probe 2"

Definition	
Type	Heat Flux
Location Method	Geometry Selection
Geometry	1 Face
Suppressed	No
Options	
Result Selection	Total
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
Maximum Value Over Time	
Total	149.59 W/m²
Minimum Value Over Time	
Total	149.59 W/m²

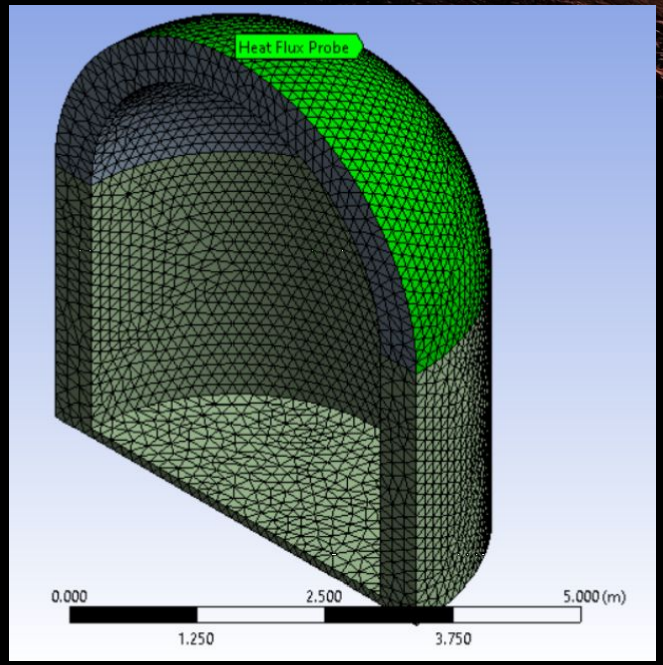
Simulation Results (Cyc2)



Details of "Temperature Probe 2"

Definition	
Type	Temperature
Location Method	Geometry Selection
Geometry	1 Face
Suppressed	No
Options	
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
Maximum Value Over Time	
Temperature	-30.431 °C
Minimum Value Over Time	
Temperature	-30.431 °C

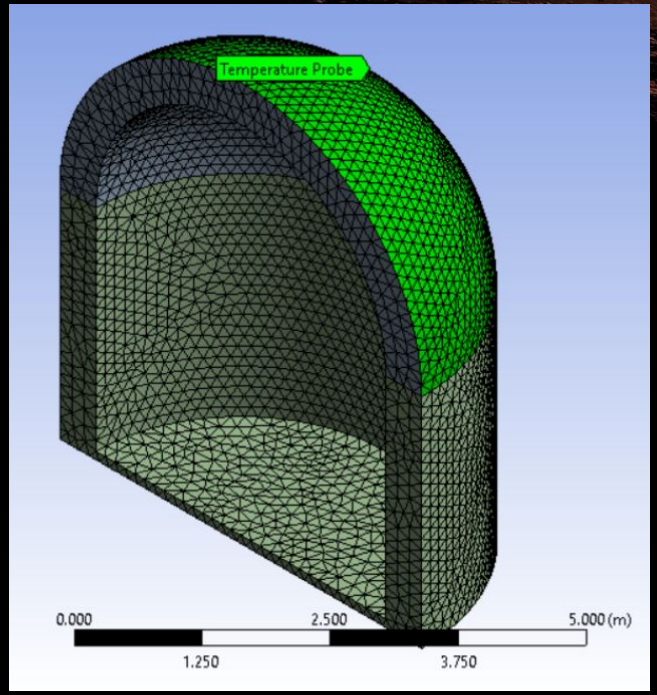
Simulations Results (Dome1)



Details of "Heat Flux Probe"

Definition	
Type	Heat Flux
Location Method	Geometry Selection
Geometry	1 Face
Suppressed	No
Options	
Result Selection	Total
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
Maximum Value Over Time	
Total	136.56 W/m²
Minimum Value Over Time	
Total	136.56 W/m²
Information	

Simulations Results (Dome2)



Details of "Temperature Probe"

Definition	
Type	Temperature
Location Method	Geometry Selection
Geometry	1 Face
Suppressed	No
Options	
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
Maximum Value Over Time	
Temperature	-20.656 °C
Minimum Value Over Time	
Temperature	-20.656 °C
Information	

Comparison

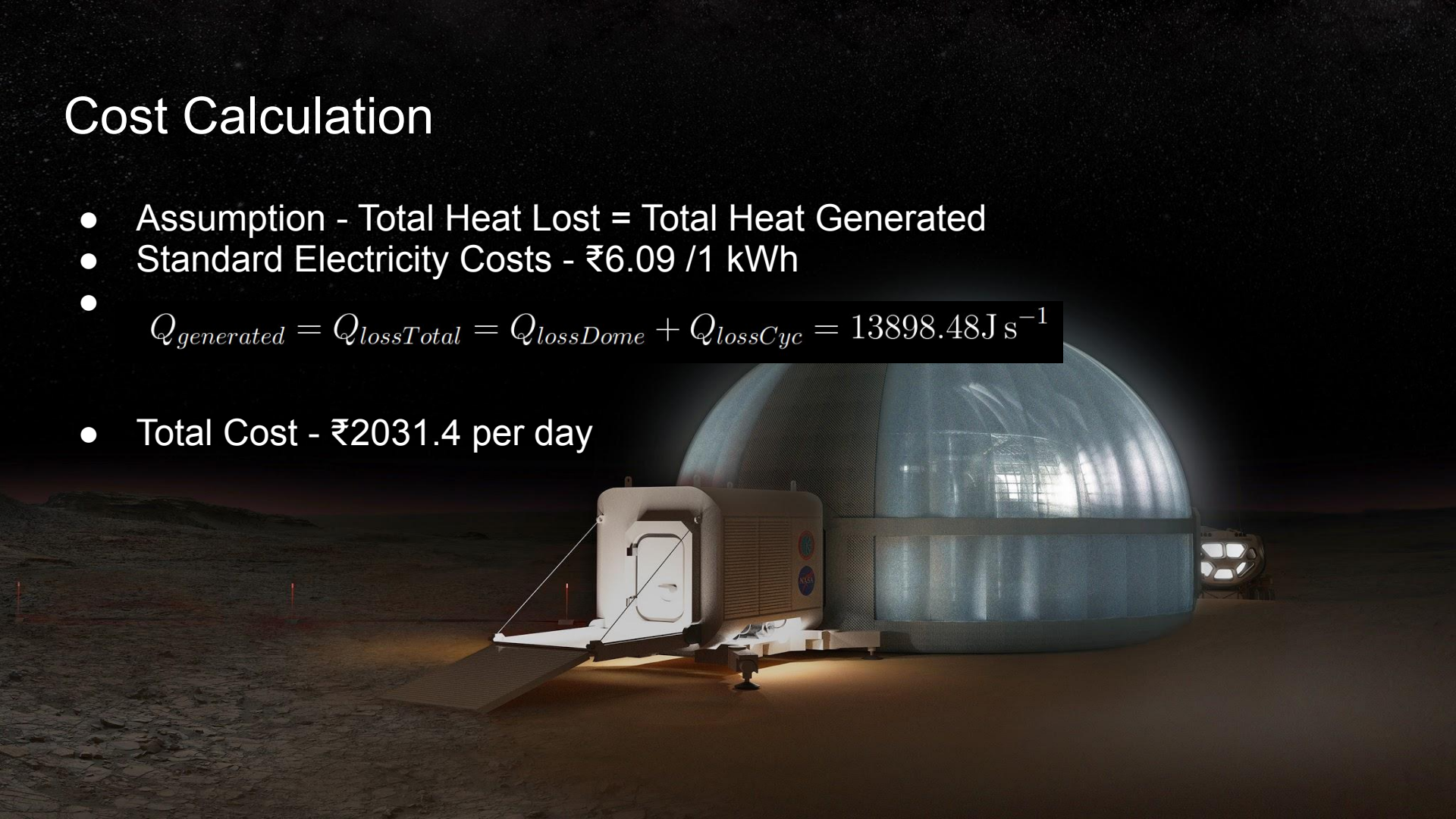
Variable	Calculated	Simulations
T_8 (Cyc)	−47.739°C	−30.431°C
T_8 (Dome)	−34.219°C	−20.656°C
q_loss (Cyc)	160.89 W / m²	149.59 W / m²
q_loss (Dome)	160.86 W / m²	136.56 W / m²

Reasons for discrepancy:-

- Simulations considered Thermal Expansion
- Simulation solved Transient Equation
- Data Extrapolation
- Nusselt Number Correlations

Cost Calculation

- Assumption - Total Heat Lost = Total Heat Generated
- Standard Electricity Costs - ₹6.09 /1 kWh
- $$Q_{generated} = Q_{lossTotal} = Q_{lossDome} + Q_{lossCyc} = 13898.48 \mathrm{J\,s^{-1}}$$
- Total Cost - ₹2031.4 per day



Conclusion

- Calculations close to simulations
- Habitable temperature maintained with reasonable costs
- Structure performed well
- Hope persists!



A Martian Sunset

Thank You!

